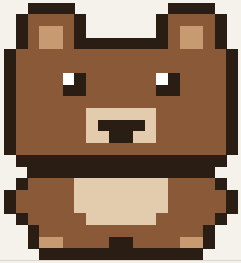
ROMA TRE · A.Y. 2026/27

Policy Evaluation Lab

Lecture 6: Difference-in-differences



Dr. Matteo Caruso · mcaruso4@os.uniroma3.it
Roma Tre University – Department of Business Economics
October, 2026

Where we are

- Lectures 4–5: when we can **randomise**, the comparison group is built for us
- Most policies are not randomised. This week and next: four designs that **find** a credible comparison group in existing data

Design	The comparison group	Key assumption
Difference-in-differences	units not (yet) treated, over time	parallel trends
Regression discontinuity	units just below a cut-off	no sorting at the cut-off
Instrumental variables	units nudged by an outside factor	the factor works only through the treatment
Synthetic control	a weighted mix of untreated units	the mix tracked the treated unit before

Today: the first row. We care about the logic and the assumption; the software does the arithmetic.

PART 1

The idea



From matching to differences

- Back to our grant (Lecture 2): firms that apply are more motivated, better managed, more ambitious
- **Matching** compares each funded firm with a similar unfunded one, but only on what we **observe** (size, sector, age)
- Motivation and management are **not** in the data, so matching cannot remove them

> **THE DIFFERENCE-IN-DIFFERENCES MOVE**
>
> If the unobserved differences between two groups are **constant over time**, comparing **changes** instead of **levels** cancels them out

- Two snapshots are not enough: we need each group **before and after** the policy

The idea is old: John Snow and cholera

- London, 1850s: is cholera carried by air or by water?
- Two water companies served neighbouring houses from the Thames, downstream of the sewers
- In 1852 Lambeth moved its intake upstream; Southwark & Vauxhall did not
- Snow compared the **change** in deaths between the 1849 and 1854 outbreaks

Water supply	Cholera deaths, 1849, rate per 100,000	Cholera deaths, 1854, rate per 100,000	Difference in rates comparing 1854 to 1849, rate per 100,000
Southwark & Vauxhall Company only	1349	1466	118
Lambeth Company Only	847	193	−653
Difference-in-difference, Lambeth versus Southwark & Vauxhall	502	1273	−771

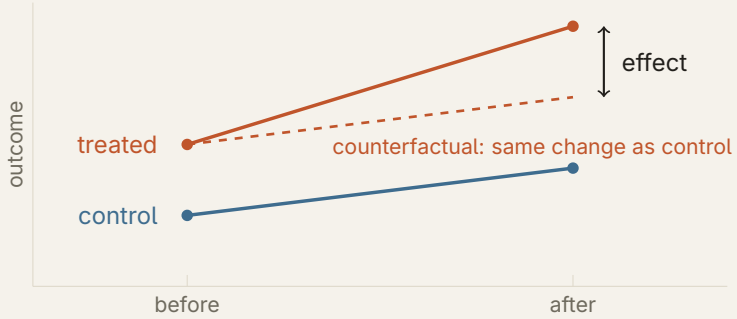
Cholera deaths per 100,000 people.

Four numbers

	Before	After	Change
Treated	$\bar{Y}_{T,\text{before}}$	$\bar{Y}_{T,\text{after}}$	Δ_T
Control	$\bar{Y}_{C,\text{before}}$	$\bar{Y}_{C,\text{after}}$	Δ_C
Difference-in-differences			$\Delta_T - \Delta_C$

- Δ_T alone (before–after) mixes the policy with everything else that changed at the same time
- Treated minus control **after** alone mixes the policy with permanent differences between the groups
- $\Delta_T - \Delta_C$ removes both, **if** the control group's change is what the treated would have experienced without the policy

The picture



Illustrative numbers. The gap at the start does not matter; what matters is that, without the policy, the two lines would have moved together.

PART 2

Parallel trends



The one assumption that matters

PARALLEL TRENDS

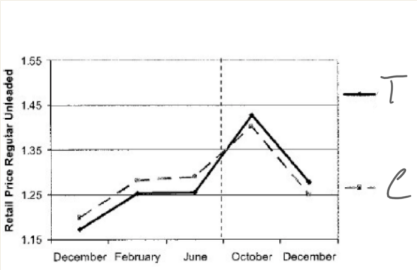
Without the policy, the outcome of the treated group would have changed by the same amount as that of the control group

- It is about the **counterfactual**, so it can never be tested directly
- The groups can differ in **levels**; they must not differ in **trends**
- It fails when something else hits one group at the same time, or when the groups were already drifting apart

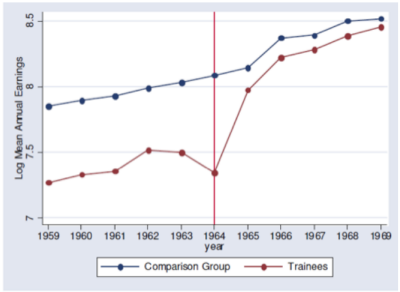
WHAT WE CAN DO

Look at **several years before** the policy. If the two groups moved together then, parallel trends after is more believable (not proven)

Checking the years before



Meeting the Parallel Trend Assumption



Violation of the Parallel Trend Assumption

With many periods we can plot the gap between the groups year by year: an **event study**. Flat before the policy, moving after: that is the pattern we hope for. Try it: EconVis, event study.

PART 3

A classic: the minimum wage



Card and Krueger (1994): does a higher minimum wage cost jobs?

- Textbook prediction: in a competitive market, a higher minimum wage reduces employment
- With employer market power, it need not: an empirical question
- **1 April 1992**: New Jersey raises its minimum wage from $4.25 to $5.05; neighbouring **Pennsylvania** does not
- Survey of **410 fast-food restaurants** in both states, before (Feb–Mar 1992) and after (Nov–Dec 1992)

> **WHY FAST FOOD?**
>
> Many minimum-wage workers, similar products and jobs across chains, no tips: restaurants on the two sides of the border are comparable

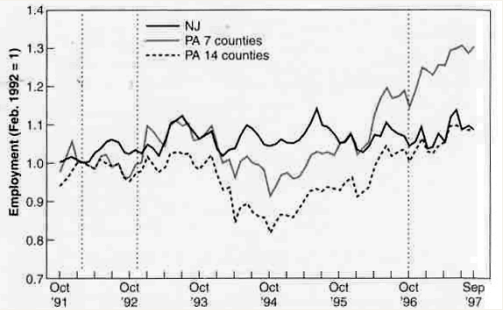
The result

FTE per store	PA	NJ	NJ – PA
Before	23.33	20.44	−2.89
After	21.17	21.03	−0.14
Change	−2.16	0.59	**2.76**

FTE = full-time equivalent employees. Card and Krueger (1994), Table 3, row 1–3.

- New Jersey stores were **smaller** to begin with: levels differ, and that is fine
- Employment fell in Pennsylvania and rose slightly in New Jersey
- DiD: **+2.76** employees per store (about 13%), significant at 5%
- The opposite of the textbook prediction

Was Pennsylvania a good counterfactual?



- With two survey waves we cannot see trends
- Later payroll data (Card and Krueger, 2000) cover 1991–1997
- Around the 1992 surveys the series move together; afterwards, employment in parts of Pennsylvania falls for years while the minimum wage is unchanged
- The debate on this study lasted years: it is a debate about **parallel trends**, not about the arithmetic

Vertical lines: the two original surveys (1992) and the October 1996 federal increase.

PART 4

Firms as targets: Law 488



Law 488/1992: capital grants for the South

- For years the main Italian instrument against territorial gaps: 1996–2003, **€16 billion** to industrial firms, 27,846 projects, mainly in southern regions
- Firms apply, are **ranked** on pre-set criteria, and are funded from the top until the regional budget runs out
- Question: did the grants create investment that would not have happened otherwise?

BRONZINI AND DE BLASIO (2006): THE DESIGN

- **Treated**: funded firms. **Control**: firms that applied but ranked too low
- Rejected applicants wanted to invest too: closer to funded firms than non-applicants (Lecture 2)
- Ministry files linked to financial statements, 1993–2001: years before **and** after

What they found, and why it matters for policy

- Funded firms **increased investment**, mainly in the second year of the grant
- In the following years their investment **slowed down** relative to rejected firms
- Interpretation: part of the subsidised investment was **brought forward** from later years, not created

> **THE POLICY LESSON**
>
> A before–after comparison, or a look at the grant years only, would have overstated the effect. The control group **and** the years after the programme changed the conclusion

Thursday: the same ranking allows a second design, regression discontinuity (Cerqua and Pellegrini, 2014, reach a more positive conclusion).

PART 5

Staggered adoption



When units are treated at different times

- Policies often roll out gradually: regions, branches, municipalities join in different years
- Each year we compare units that have just joined with units that have not (yet)

$$Y_{it} = \underbrace{\mu_i}_{\text{unit}} + \underbrace{\tau_t}_{\text{year}} + \gamma\, D_{it} + \varepsilon_{it}$$

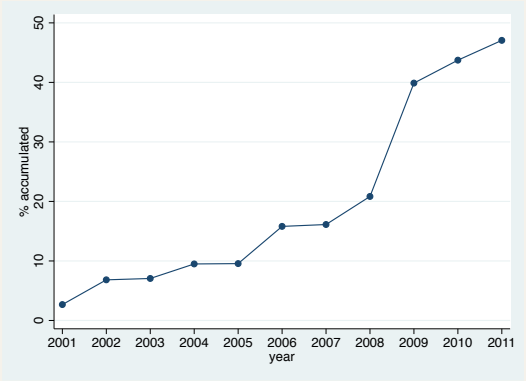
μ_i: permanent differences between units; τ_t: shocks common to all in a year; γ: the DiD effect

A WARNING

If the effect grows or fades over time, this regression can use early adopters as controls for late ones. Newer methods compare only with not-yet-treated units (Goodman-Bacon, 2021)

Municipal unions in Emilia-Romagna

- Ferraresi, Migali and Rizzo (2018): do small municipalities save money by sharing services in a **union**?
- Emilia-Romagna, 2001–2011; municipalities join unions in different years
- Outcome: current spending per inhabitant



Share of municipalities in a union.

Municipal unions: results

	Full sample		
	(1)	(2)	(3)
Municipal Union	-0.065***	-0.047***	-0.030**
	(0.012)	(0.015)	(0.015)
Permanence			-0.022***
			(0.006)
Permanence square			0.002***
			(0.000)
N	3686	3586	3586
Year FE	✓	✓	✓
Municipality FE	✓	✓	✓
Municipality controls		✓	✓

- Joining a union lowers current spending per inhabitant by about **3 to 6%**, depending on the controls
- The estimate shrinks with more controls: what else differs between joiners and non-joiners?
- Efficiency or fewer services? The authors argue services did not fall; another study of Italian unions finds no clear effect (Luca and Modrego, 2021)

Coefficients on log spending; municipality and year fixed effects in all columns.

PART 6

Reading a DiD study



Five questions to ask any DiD study

1. **Who is the control group**, and why would it have followed the same path?
2. Are there **several years before** the policy, and do the groups move together in them?
3. Did anything **else** hit the treated group at the same time (another reform, a local shock)?
4. Did the policy change **who is in** the groups (firms moving, closing, entering)?
5. Do the results survive a **different control group** or a different period?

> Strength: credible effects from data that already exist. Weakness: everything rests on one assumption about a path we never observe

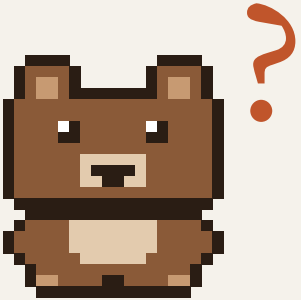
Friday lab and readings

- **Friday**: two studies side by side (Card and Krueger; Bronzini and de Blasio). For each: the five questions, then what a policymaker should take from it
- Interactive: EconVis, difference-in-differences and event study (Exercises page)
- Readings on the Lectures page, week 4

> **THURSDAY**
>
> Regression discontinuity: what a funding cut-off can tell us about R&D grants

References

- Bronzini, R. and G. de Blasio (2006), Evaluating the impact of investment incentives: the case of Italy's Law 488/1992, *Journal of Urban Economics*, 60(2), 327–349
- Card, D. and A. B. Krueger (1994), Minimum wages and employment: a case study of the fast-food industry in New Jersey and Pennsylvania, *American Economic Review*, 84(4), 772–793
- Card, D. and A. B. Krueger (2000), Minimum wages and employment: a case study of the fast-food industry in New Jersey and Pennsylvania: reply, *American Economic Review*, 90(5), 1397–1420
- Cerqua, A. and G. Pellegrini (2014), Do subsidies to private capital boost firms' growth? A multiple regression discontinuity design approach, *Journal of Public Economics*, 109, 114–126
- Ferraresi, M., G. Migali and L. Rizzo (2018), Does intermunicipal cooperation promote efficiency gains? Evidence from Italian municipal unions, *Journal of Regional Science*, 58(5), 1017–1044
- Luca, D. and F. Modrego (2021), Stronger together? Assessing the causal effect of inter-municipal cooperation on the efficiency of small Italian municipalities, *Journal of Regional Science*, 61(1), 261–293
- Goodman-Bacon, A. (2021), Difference-in-differences with variation in treatment timing, *Journal of Econometrics*, 225(2), 254–277
- Snow, J. (1855), *On the Mode of Communication of Cholera*, 2nd edition, London: John Churchill



Questions?

mcaruso4@os.uniroma3.it · https://pel.carusomatteo.it